

A PRACTICAL BUILDER'S GUIDE

BUILDING SPECIALTY PACKS

A Practical Guide to Constituting Domain Reality
for Temporal Standing



TIM ZLOMKE

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Copyright and edition notes

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This practitioner guide develops consequence-specific domain constitution. The companion project contains reference configurations, source lineage, and examination evidence. Examples use synthetic institutions and do not prescribe professional practice in any particular jurisdiction or domain.

The pack constitutes. Harmonic determines. The executor enforces.

Source claims, standing determinations, and downstream effects have distinct evidence boundaries. Repository-backed results are local examinations of the identified snapshot. The source and evidence notes explain the scope of each result.

Read the chapter prose for the method. Use each builder contract to identify the concrete artifact and its falsifier. Use the companion project for scripts, packets, and retained examination reports.

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A Note from the Author

I wrote this book for the people who will build what comes next.

As our systems become more capable of acting, we owe the people affected a clear account of the grounds on which we proceed. A genuine approval can cease to support a later action. An accurate record of yesterday can be insufficient for today. Builders must preserve those distinctions.

Does It Still Stand? explored the question of temporal standing. This book takes up the next responsibility: establishing what an acting system may legitimately reason over. A Specialty Pack makes that responsibility concrete through declared sources, scoped evidence, domain meaning, and inspectable relationships.

I want you to understand the method, challenge its assumptions, and examine the evidence. The companion contains versioned source, synthetic examples, and bounded examination records. Tested claims should have inspectable records. Work that remains should stay visible.

The examples are starting points. Your institution must establish its own governing rules, competent sources, and consequence boundaries. Carry the method forward with care. Do not let a convenient default become an invented fact, or let an earlier permission quietly become permanent authority.

For me, stewardship means accepting responsibility for the systems we enable and for the people who must live with their consequences. Clarity is part of that responsibility. So is leaving a record that another builder can inspect, reproduce, and improve.

I hope this becomes a reference you return to when sources disagree, conditions change, or the next action matters.

Build carefully. Make the grounds visible. Give the next person something they can trust enough to examine.

Tim Zlomke

Founder, Moral Clarity AI

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How to Use This Book

This book is for builders who need an acting system to use institutionally justified evidence before producing a particular effect. It assumes no prior reading of *Does It Still Stand?* Start with Chapters 1-4, then work through the construction and examination method before adapting a domain example.

The question in ordinary language

Suppose a worker scheduled an approved promotional message yesterday. Today the recipient withdraws permission. The old approval remains a real historical record. The practical question is whether the grounds for this particular send remain operative when the worker attempts it.

The institution declares which sources and rules establish the relevant grounds. The pack assembles attributable evidence and interprets its domain relationships. Harmonic evaluates standing for the proposed consequence. The execution boundary applies that determination at the component capable of producing the effect. A successful assembly can describe a withdrawal just as completely as an approval.

Terms you will need

- **Institution:** the organization or governing arrangement responsible for declaring the relevant rules and source responsibilities. Naming an institution does not establish its legitimacy.
- **Consequence:** the particular effect being proposed, including the identities and scope that materially distinguish it from another effect.
- **Proposition:** a claim that can be evaluated, such as whether this recipient is suppressed for this purpose.
- **Predicate:** a condition expressed over evidence. Its adequacy depends on the institution's declared consequence and rules.
- **Source competence:** the declared responsibility of a source to establish a particular proposition within a scope. A source's popularity or recent timestamp does not supply that responsibility.

- **Constitution:** assembling and interpreting the evidence that forms the declared domain state. In this book, this is a technical responsibility; it does not itself grant execution permission.
- **Standing:** whether the governing grounds support continuation for the proposed consequence at the relevant time. Concrete runtime outcomes and execution fields must be interpreted under their actual interface contract.
- **Principal:** the person or entity on whose behalf an action is proposed. Delegation establishes a scoped relationship between that principal and an actor.
- **Falsifier:** an observation that would show a stated claim failed. Each chapter gives one so a builder can challenge the method rather than simply copy it.

Follow one send

Message M17 is the proposed content, campaign C4 is its campaign, and recipient R1 is its intended recipient. These identifiers name synthetic examples, not actual people or accounts. A preference registry supplies scoped preference and suppression observations. A delegation service supplies the sender's authority relationship. A content repository supplies the approved content version. A quota ledger supplies a reservation claim for the particular send.

If the registry supplies a current withdrawal, preserve it. If it cannot supply the required observation, report the missing evidence. Those are different states. Neither state should be rewritten into a favorable default. Only after the required domain and governing information is constituted can the evaluator perform its responsibility.

Read claims at their proper level

The architectural boundaries are the design advocated by this book. The domain configurations are illustrative institutional choices. Observations about Harmonic refer to the supplied versioned source snapshot. Local test results establish only the behavior exercised by those tests. The source and evidence notes identify those boundaries and the companion records to inspect.

The six domain examples are starting points for institution-specific design. Their rules, source hierarchies, freshness windows, and expected outcomes are not universal requirements. Before using one, assign competent sources, review consequence coverage with domain owners, and examine the actual acquisition, evaluator, and execution path.

Define What Must Be Governed

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 1 - What a Specialty Pack Constitutes

The first question was: Does it still stand? The next question arrived almost immediately: According to what?

An acting system encounters records, services, approvals, identities, policies, contracts, preferences, and events. It does not encounter reality in an unmediated form. Somebody must establish which sources matter, what each source is competent to establish, and what happens when their observations disagree. That responsibility belongs to the institution. A Specialty Pack makes the institution's constitution rules inspectable and executable.

Consider an outreach worker. For this example, the institution designates the CRM to establish contact affiliation, the preference registry to establish suppression, the delegation service to establish who may send on whose behalf, and the content repository to establish the approved message version. Each source has a declared responsibility; none is assumed infallible. None of those sources individually supplies present standing for the exact send. The pack constitutes their relevant propositions; Harmonic determines standing; the boundary enforces the determination.

This division does not make the pack a passive folder of documents. It contains meaningful interpretation: source scope, precedence, event semantics, and evidence requirements. But it does not get to insert the answer for the current case. A configuration that says 'permit this customer' has crossed the boundary unless it is recording an actual, scoped institutional authority that the case must still be evaluated against.

A Specialty Pack does not tell Harmonic what answer to give. It tells Harmonic what reality it is legitimately allowed to reason over. Begin by drawing the responsibility map. If two components both claim to decide standing, or no component owns source competence, fix that architecture before collecting more data.

Specialty Pack and Governance Pack are different responsibilities

The supplied repository uses Governance Pack for Harmonic's governance pipeline. This book uses Specialty Pack for institutionally constituted domain meaning. A communications Specialty Pack is not a second implementation of the runtime pipeline.

The V112 freeze expresses the boundary precisely: the Specialty Pack establishes the attributable domain relationship; Harmonic determines its consequence-relative constitutional significance. V113 then preserves a discrimination pair after the interface receives complete governing information. That lineage is stronger than a general description because its proposition and boundary are inspectable.

Avoiding answer leakage must not make the packet so sparse that the runtime cannot evaluate its legitimate responsibility. An implementation can fail by supplying the answer. It can also fail by withholding the required governing basis. Both failures belong in examination.

Builder contract

- Purpose: Establish the institutional, pack, Harmonic, and executor boundaries.
- Central claim: Constitution and standing determination have distinct owners.
- Falsifier: A pack supplies a case-specific execution answer, or Harmonic invents upstream authority.
- Implementation artifact: Responsibility map and boundary contract
- Concrete example: An outreach worker combines four sources before one send.

Chapter 2 - Start With the Consequence

A consequence definition is the first configuration file. Everything else depends on it.

'Send email' is an operation name. It does not tell a reviewer who acts, who receives the message, which purpose applies, or what will actually be transmitted. Define the effect as a tuple: acting and represented principal, recipient identity, purpose, message version, material attachment identities, campaign, channel, provider, and execution instance. Include only fields whose alteration could change the governing basis or effect identity.

Do not confuse a contact identifier with a verified delivery destination. A CRM contact can acquire a new address. Nor does a content identifier necessarily identify the final message: a renderer may add an attachment after determination. Trace the proposed payload through its last material transformation. Decide which representation is evaluated and which representation reaches the provider.

The practical examination is substitution. Replace the recipient. Change the purpose from a service notice to promotion. Change the attachment. Ask whether the old identity still matches. If a material substitution is invisible, the definition is too coarse. Conversely, changing a nonmaterial internal trace label should not force a new governing case.

Canonicalization is an engineering specification, not a decorative hash. Define how strings, numbers, absent fields, array order, and versions are represented. A digest can bind the selected representation; it cannot correct the omission of a material field. Start with a worksheet, then implement the fingerprint only after the material-field choices are reviewable.

Consequence worksheet: one message, three identities

An outreach system commonly has a contact identity, a delivery destination, and a send instance. Record all three. Contact R1 establishes who the institution means to reach. Address A1 establishes where the provider will deliver. Send S1 distinguishes one proposed consequence from another. A changed address may preserve the contact while changing the delivery consequence. A repeated send may preserve both while creating another effect.

Now add payload and purpose. Message M17 contains a brochure approved for a named campaign. A service notice using the same address may depend on a different governing basis from promotion. The purpose is therefore a material field, even though the provider accepts the same HTTP operation.

Follow rendering to its final material form. If personalization changes only an approved salutation under a bounded content rule, the institution may govern that transformation through the rule. If it introduces a new claim or attachment, the change may require another content observation. Document the distinction; do not leave it to the renderer's confidence.

Define identity before choosing a digest

Use stable identifiers for institutions, principals, targets, and artifacts. Define whether omitted, empty, and null fields differ. Preserve numeric meaning for amounts. Declare whether ordered attachments have a material order. Then compare the actual provider object with the evaluated representation.

The local examples serialize selected JSON fields before hashing. Their digest is useful for detecting represented changes. It does not establish a universal cross-language canonicalization profile or prove that every material provider field was included. Examine that profile when another language or connector is introduced.

The acceptance exercise is concrete: give a reviewer two payloads and ask which differences alter the governing case. Every such difference should appear in the consequence definition or an explicitly governed transformation. The artifact is complete enough to implement when the

reviewer can explain why each material substitution does or does not require a new evaluation.

Builder contract

- Purpose: Specify the exact external effect.
- Central claim: Materially different consequences are distinguishable.
- Falsifier: Recipient, purpose, or payload substitution retains the same governed identity.
- Implementation artifact: Consequence worksheet and identity specification
- Concrete example: M17 to R1 differs from M17 to R2.

Chapter 3 - Build the Standing Predicate Set

A predicate earns its place by answering a consequence-relative question. Data availability is not a sufficient reason to collect it.

For an outreach send, the institution may require recipient identity, communication preference, suppression state, campaign approval, content approval, sender delegation, applicable account state, reply state, and a quota allocation. Record what proposition each predicate means. 'Quota available' is ambiguous if it means an observation made before ten other workers consumed the same allowance. A reserved allocation and an observed counter are different propositions.

Book 1's authority, identity, scope, state, policy, evidence, and consequence categories provide a checklist, not an automatic complete inventory. Walk the effect with domain owners. Ask what condition would make this exact effect unjustified, what uncertainty would prevent deciding, and what benign changes should leave it unaffected. Preserve the rationale for including or excluding each candidate.

Classify dependencies prospectively. Required evidence must not become optional because its service is down. Optional context must not quietly acquire veto power because it happens to be present. Conditional requirements need explicit activation rules: an attachment-approval predicate matters when an attachment is proposed.

The register is a bounded institutional declaration. A structural test can establish that every declared predicate has a source and freshness rule. It cannot establish that the institution remembered every material condition. That second claim needs domain examination and counterexamples. Keep the distinction in the release report.

Domain propositions are not automatically runtime witnesses

The six synthetic proposition registers are inspectable. They do not establish a universal production mapping from each boolean into a Harmonic field. The inspected interface also receives structured understanding, authority, obligation, provenance, and consequence witnesses, with their own meanings and attribution.

A complete local register does not automatically establish current authority, intact historical provenance, satisfied obligations, or coherent understanding. Record the proposition each runtime field represents, what establishes it, and the transformation used. Setting every witness true after checking JSON syntax promotes formatting success into governing substance.

An explicit not-applicable obligation witness is different from failing to collect obligations. The former is an attributable applicability claim; the latter is an evidence gap. Likewise, an empty revocation list needs a legitimate acquisition and freshness basis. An array created because no service was consulted establishes nothing about revocation.

Builder contract

- Purpose: Derive required propositions from the effect.
- Central claim: Every predicate has a relevance rationale and declared evidence requirement.
- Falsifier: A known material condition is omitted or an unrelated predicate is collected without justification.
- Implementation artifact: Predicate register
- Concrete example: Quota reservation is distinct from reading a counter.

Establish What Sources May Govern

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 4 - Map Sources to Propositions

A source does not become universally authoritative because the institution calls it a system of record.

Map authority at the proposition level. The CRM may establish account ownership while the preference registry establishes suppression. The identity service may establish a principal's identity while a separate delegation instrument establishes what the principal may do. An accurate summary of all three remains a summary, not the authority that created any of them.

Each mapping needs an owner, authority basis, permitted proposition scope, effective interval or applicability conditions, and change mechanism. Distinguish the authoritative system from the adapter that reads it. An adapter version matters because it can misinterpret a status even when the underlying record is correct. Preserve both identifiers when interpretation depends on the adapter.

Source identifiers must not be accepted on self-assertion alone in a production integration. A packet saying 'source: consent-registry' proves only that somebody wrote those characters. The institution must establish how the claim is acquired, authenticated, and connected to the legitimate source. The companion's synthetic registry assumes that trust relationship; it does not demonstrate it.

Use a negative-scope column. 'CRM establishes relationship; CRM does not establish promotional preference' is easier to enforce than a vague instruction to trust CRM. A useful examination substitutes a plausible but incompetent source. The resulting evidence should remain insufficient for that proposition even when it is recent and well formed.

Establishment needs evidence behind the label

The interface includes `present_state_provenance` and an epistemic status vocabulary. ESTABLISHED records a caller's attributable representation; it does not independently prove the proposition.

Identify who established the state, when, from which evidence, through which method, and within what scope. Derived relationships need source observations, rule identity and version, derivation lineage, and consequence scope. A confidence score cannot replace source competence.

V113 retains attributable facts, a frozen relational rule, the pack relationship, exact request, and response. That is the pattern to learn from. It does not establish independent truth of future arbitrary facts or authorize an adapter to stamp new facts as established without legitimate institutional work.

Builder contract

- Purpose: Declare source competence for each proposition.
- Central claim: Authority mappings are attributable and scope-specific.
- Falsifier: A recent CRM note is accepted as preference authority without a legitimate mapping.
- Implementation artifact: Source register
- Concrete example: CRM relationship evidence cannot establish suppression.

Chapter 5 - Declare Precedence and Preserve Conflict

Conflict is information. Flattening it can manufacture a certainty no source supplied.

Suppose an operative signed restriction and a newer informal note disagree. The newer timestamp is not a governing rule. The institution must declare which source controls this proposition, in this scope, under these conditions. A hierarchy may be partial rather than a universal ranking: one source controls employment status, another communication preference, and a competent override may apply only to a narrow exception.

Record the conflicting claims before recording the resolution. A precedence record should identify the applicable rule, its version, the selected proposition, and why its scope covers the conflict. Retaining a losing claim does not make it operative. It preserves the reason the institution did not rely on it.

When no declared rule resolves disagreement, preserve unresolved conflict. An operator can seek clarification from the competent authority, but an urgent workflow does not create that competence. A clarification should become attributable evidence with its own scope and version, not a hidden edit to the packet.

Examine precedence by reversing observation order. The result should not change merely because the weaker source arrived last. Then remove the precedence rule. The same claims should now remain unresolved. This tests declared semantics rather than a developer's accidental sorting behavior.

A precedence matrix should have a reason column

For each proposition, identify the controlling source, any competent subordinate source, the scope of its precedence, and the institutional rule establishing that relationship. A rank without a reason is just another hidden preference.

Consider a primary authority register and a delegated register for execution region. Both may be competent to supply region claims under an illustrative constitution. The primary source controls in a disagreement because the declared rule gives it that role. A newer delegated observation does not outrank it by being newer.

The relationship example preserves all examined claims in its resolution record. It names the proposition, rule version, controlling sources, and weaker conflicting evidence. This prevents the final selected value from erasing how disagreement was handled. The record does not declare overall standing; it establishes the domain proposition used in the comparison.

Know when the hierarchy cannot help

Two equal-rank claims disagreeing at the controlling level remain unresolved in the example. It does not select the latest timestamp because no version-order or supersession rule has been established. If a real source supplies authoritative version ordering, the institution may define that interpretation explicitly and examine it separately.

Stale controlling evidence also remains unresolved. A weaker current record does not silently become controlling because the primary source is inconvenient. A legitimate fallback requires its own scope, activation conditions, source competence, and examination evidence.

The example deliberately holds when a submitted candidate claim is malformed or stale rather than silently discarding it. That is an illustrative constitution choice. Another institution may define an admissible-evidence selection rule differently, but it must record why an excluded claim is not required and whether its exclusion creates uncertainty for the consequence.

Test the same claims in reversed arrival order. A declared hierarchy should retain the same governing proposition and preserve equivalent resolution evidence. If the answer changes only because the array order changes, the software has implemented accidental precedence.

Builder contract

- Purpose: Represent disagreement and legitimate resolution.

- Central claim: Only applicable precedence or competent clarification resolves conflict.
- Falsifier: Arrival order silently determines which claim controls.
- Implementation artifact: Precedence matrix and conflict record
- Concrete example: An operative restriction controls over a recent informal note.

Chapter 6 - Define Freshness by Consequence

Freshness answers whether this observation is usable for this consequence. Authority answers whose observation matters. Neither substitutes for the other.

A durable delegation might remain operative until revocation within an explicit validity interval. A rapidly changing suppression source may need a short observation window, reliable invalidation events, or a version check at the execution boundary. The institution chooses these requirements according to the consequence and source behavior. The book supplies no universal timeout.

Define event-driven invalidation and an uncertainty backstop together. If the event stream is healthy, a relevant revocation invalidates the affected state. If event delivery becomes uncertain, the maximum tolerated observation age prevents silence from looking like continuity forever. Specify how health is established; the absence of messages is not automatically proof of a healthy stream.

Do not turn freshness into repeated ceremony. Book 1's durable-permission counterexample matters: justified continuity can support useful automation. A benign profile change should not invalidate a narrowly scoped permission to retrieve invoice metadata.

Test the edges. Inspect observations exactly at the age limit, beyond it, and unexpectedly in the future. Declare clock assumptions and permitted skew. Then test a revocation that arrives inside the nominal cache interval. If the timer defeats the invalidation signal, the implementation has misunderstood its own freshness contract.

Write a freshness rule as a procedure

'Current enough' needs an operational definition. For one suppression proposition, a rule could require a source observation no older than a declared interval, no known later revocation event, and an established source-health condition. These are example rule dimensions, not recommended default intervals.

Specify the comparison instant. A worker may gather evidence, wait in a queue, and dispatch later. A pack check at assembly does not prove the observation remains usable at dispatch. State which component rechecks age or versions and which signals invalidate the assembled state.

For a durable grant, the procedure may differ: verify its effective interval, scoped applicability, revocation observation, and required source-health evidence. A new human approval is unnecessary where declared continuity remains justified. The freshness matrix should make that positive path as explicit as the stop conditions.

Outages need evidence semantics

If the primary service is unavailable, the cache contains historical evidence. Whether it remains usable depends on the declared rule. A cache hit does not establish service health. An empty response does not establish no revocation. A timeout does not establish authority was withdrawn.

Keep the reason precise. Stale evidence requires refresh or an authorized alternative. Unknown source health may require restoring the change channel. Actual revocation requires a new legitimate basis before the affected consequence can proceed. Sending every case through a generic retry loop conceals these differences.

The synthetic examiner fixes time for reproducibility. That makes the fixture age calculation inspectable. It is not a pattern for a production clock. An execution system must establish its current time basis and examine clock skew, delayed jobs, and the interval between evidence acquisition and effect.

Builder contract

- Purpose: Specify usable observation age and invalidation.
- Central claim: Freshness is explicit for each required proposition and consequence.
- Falsifier: A declared revocation or expired observation remains usable.
- Implementation artifact: Freshness matrix

- Concrete example: A durable grant and suppression observation have different rules.

Chapter 7 - Establish the Historical Governing Basis

Authenticity is necessary for many evidence claims. It is insufficient to establish that a collection of records once operated together.

Imagine an approval covering message M16 and a recipient authorization established later. Both records are real. Their timestamps are correct. Nothing proves that sending M17 to that recipient was ever supported by their joint basis. A retrospective assembly can produce a coherent-looking story while manufacturing the very T_0 it claims to preserve.

Record the constituted antecedent basis when it becomes operative where practical. Identify the consequence scope, constituent records, effective applicability, constitution rules, and basis identity. If reconstruction is necessary, distinguish reconstructed lineage from a contemporaneously recorded basis. Identify the evidence that establishes joint applicability rather than relying on chronology alone.

Overlapping dates are helpful but not conclusive. Records may refer to different purposes, different targets, or incompatible policy versions. Conversely, an institution may legitimately use a durable grant whose applicability persists across many observations. The task is to establish its actual governing relationship, not to demand that every record share one timestamp.

An unresolved antecedent basis should remain visible. Later authorization can support a new basis if legitimately established; it does not retroactively repair an unsupported historical account. The pack must preserve both what is known and what is merely reconstructed. Do not let elegant lineage formatting do the work of evidence.

Worked basis: two genuine approvals, no joint authority

On Monday, an institution approves message M16 for campaign C4. On Tuesday, a recipient's preference covers a promotional send under C4. On Wednesday, a worker renders M17 with a new attachment and assembles the Monday approval and Tuesday preference. Every record is authentic. The assembled case still lacks evidence that content authority covers M17.

The basis record should identify the exact propositions the instruments establish. Content approval covers a named version and its scope. Preference covers a recipient, channel, purpose, and applicable conditions. The constitution rule establishes how those propositions may support one proposed consequence. If a material component is absent, record that dependency rather than relabeling the nearest historical record.

Consider a less obvious variation. Content approval covers M17, but ends at noon Tuesday. Recipient authorization begins at one Tuesday. The records are chronologically ordered and individually operative at different times. They do not establish a jointly operative historical basis for the send. Effective-interval compatibility is an examination target, not a conclusion to infer from a sorted event list.

Contemporaneous record versus later reconstruction

Where possible, retain a basis identity when the institution establishes the governing combination. A useful record contains the constituent instrument identities, their versions, consequence scope, applicability evidence, governing rule version, constitution time, and responsible authority. The record's existence does not prove its legitimacy; its supporting relationship still needs examination.

Reconstruction is legitimate work when the original record is fragmented. Mark it as reconstruction. Preserve the evidence supporting joint applicability and the uncertainty that remains. A later analyst must not improve the historical basis by adding a rule that did not apply then. Nor should a later signature silently be attributed to an earlier interval.

There are three different questions: what records existed; what basis legitimately operated; what can support a consequence now. Answer each

separately. A new competent approval can establish present grounds without proving an older basis existed. A historical basis can be established without remaining operative now.

Builder procedure

Name the effect first. List each constituent authority or condition and the proposition it established. Record effective applicability, not only creation time. Check joint scope and interval compatibility using the institution's declared rules. Preserve supersession and revocation. Identify the evidence establishing the basis as a combination. If that evidence is absent, expose the gap.

The first examination should deliberately provide authentic but incompatible fragments. The second should supply a durable grant plus current conditions that legitimately preserve it. This pair prevents a validator from succeeding merely by rejecting every reconstruction.

The current executable examples do not prove historical operativity. Their `antecedent_basis` object is a synthetic record retained for inspection. The release report must keep that limitation visible until a separate basis examination is implemented.

Builder contract

- Purpose: Distinguish authentic fragments from an operative T0.
- Central claim: Joint historical applicability requires evidence of the governing relationship.
- Falsifier: Real records with incompatible scopes are reconstructed as one operative basis.
- Implementation artifact: Antecedent-basis record
- Concrete example: M16 approval does not retrospectively approve M17.

Constitute Present Domain State

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 8 - Separate Events, Transitions, and Standing

An event is a claim about change. Transition semantics explain what that claim means for a domain proposition. Standing belongs to the evaluation of the exact consequence.

CRM_CONTACT_UPDATED illustrates the problem. It could describe a spelling correction, a new employer, a changed address, or a note alleging withdrawal. The event name alone does not establish which proposition changed, whether the emitter is competent to establish it, or whether the change matters to this send.

The transition catalog links an attributable event to affected predicates and interpretation rules. A verified preference withdrawal may establish that promotional permission no longer applies. An address change may invalidate identity evidence until the new destination is verified. Loss of source access can reduce evidence sufficiency without proving revocation.

Use labels that identify predicate effects: withdrawal, scope change, supersession, source invalidation, restoration, and evidence insufficiency. If 'defeating' appears in a domain catalog, scope it to the proposition and rule. It must not mean 'the pack has already decided the entire case.' An alternate emergency authority may govern the consequence through a different legitimate basis.

Duplicate, delayed, and out-of-order events need explicit handling. A delayed grant must not erase a later revocation because it arrived last. Retain source ordering or version evidence where available. Where ordering cannot be established, represent that dependency rather than inventing an event sequence.

Relationship vocabulary is an interface commitment

In V113, the domain rule compares authority A's execution region with authority B's accepted region. Matching facts produce `consistent_with_prior_state`. Differing facts produce `material_contradiction`. Valid authority and the requested consequence remain fixed.

These labels have governing relevance without being case dispositions `PERMITTED` or `REFUSED`. Their legitimacy depends on a prospectively constituted rule, attributable facts, declared vocabulary, and scoped relationship. Choosing `material_contradiction` because a test expects refusal would be leakage. Review the derivation, not merely the output spelling.

The supplied OpenAPI does not perfectly align with the preserved accepted packets. It lists a narrower preserving vocabulary and requires fields those packets omit. The runtime replay accepts the historical representation. Retain the distinction among schema documentation, implementation behavior, and frozen evidence. Resolve a current adapter contract through a successor examination rather than normalizing the historical record.

Builder contract

- Purpose: Interpret change without replacing determination.
- Central claim: Events acquire meaning through scoped domain interpretation.
- Falsifier: A raw event label directly decides the execution outcome.
- Implementation artifact: Transition catalog
- Concrete example: A changed address invalidates destination evidence.

Chapter 9 - Assemble the Constituted Packet

Packet assembly is where fragmented reality becomes one bounded examination object. A clean JSON object is not evidence that reality was clean.

Include the exact consequence, pack identity and version, applicable institutional rule references, antecedent basis when required, authoritative propositions, evidence references, source versions, observation times, interpretation versions, relevant changes, conflicts, and missing dependencies. Include the evidence needed to understand limitations, not merely the values that make the case convenient.

Distributed reads can each be fresh without describing a coherent state. Suppose campaign approval is read before suspension, while delegation is read after a related authority update. If their joint applicability matters, the pack needs a declared consistency method: shared snapshot, compatible versions, causal markers, or an explicit account of the remaining uncertainty. Timestamps alone do not establish compatibility.

Minimize sensitive content. A reference and version may be sufficient for a proposition if the downstream determination can legitimately interpret and inspect it. A reference that nobody can resolve is not reconstructability. Separate restricted evidence from decision metadata and define access and retention.

The sample packet in this project is a conceptual model. It is not a production API declaration. Its purpose is to make omitted provenance, undeclared propositions, missing evidence, and answer leakage inspectable. A real adapter must preserve these semantics while conforming to the actual Harmonic contract.

Worked assembly: facts read at different moments

At 10:00:00, the pack reads campaign approval version C7. At 10:00:01, the campaign is suspended and the delegation service publishes an associated restriction. At 10:00:02, the pack reads delegation version D8. Both observations are younger than their maximum age. Their compatibility remains a separate question.

A useful packet records source versions and any available relationship between them. If a shared transaction snapshot is available, retain its identity. If the institution relies on causal revision markers, retain the relevant markers and declared compatibility rule. If it accepts bounded independent observations for this consequence, state that assumption and examine its effect. Freshness does not establish an atomic view of distributed reality.

Do not make packet completeness cosmetic

Begin with the prospective requirement register. Assemble observations against those requirements. Keep a finding for each missing, malformed, stale, conflicting, or out-of-scope observation. Do not create requirements by looking at what the services happened to return; that would make every successful response look complete by construction.

Normalization must preserve meaningful distinctions. Missing and false are different. No source response and an authoritative statement of no active hold are different. Unknown order state and cancelled order state are different. A convenient boolean default can erase the reason the system must remain unresolved.

The synthetic examples deliberately use typed booleans for declared propositions. This choice makes their tests small and inspectable. Real packs may require amounts, intervals, qualified identities, structured restrictions, or collections. Preserve those types when a boolean would discard scope. A proposition such as 'amount within grant' still needs an attributable relationship to the actual amount and grant.

Evidence scope and effect identity

The companion validator uses a consequence fingerprint to each synthetic evidence record. The validator rejects reuse when the packet's material consequence changes but the evidence scope fingerprint does not. This demonstrates one narrow property: the represented observation is declared for the represented effect.

It does not prove that an authority signed the fingerprint or that the underlying fact is true. Anyone editing a synthetic packet can recompute the digest. Production binding needs an established acquisition and integrity mechanism. This demonstration makes scope drift observable without pretending to implement that mechanism.

Retain enough, expose only what is needed

Evidence retention and evidence transmission are different decisions. A determination may require a minimized proposition and attributable reference, while an authorized reviewer may need access to a restricted underlying instrument. Define who can resolve references, for how long, under which access rules, and what happens when a source deletes a referenced record.

An examination bundle should contain enough to reconstruct the claim without exposing unrelated sensitive content. If reconstructability depends on a live source, record that dependency. A hash of an unavailable document establishes neither its contents nor its applicability. The objective is inspectable governing grounds, not the largest possible archive.

A packet with an examined derivation

The relationship reference compares two typed region facts under a prospectively declared equality rule. Its output includes the source observations, selected domain values, precedence records, rule identity, scope, and derivation hash. The relationship is derived from facts; there is no lookup of the expected runtime result.

The handoff assembler retains explicitly supplied institutional understanding, authority, obligation, revocation, and consequence witnesses. It does not infer them from equality. These fixed synthetic

witnesses form the successor's examination scaffold, not a source of present real-world authority.

Two local handoffs call the supplied Harmonic snapshot. Equal values produce a consistent relationship and a PERMITTED result. Differing values produce a material contradiction and a REFUSED result. The observed results are kept separate from the pack's output. This examines one attributed domain comparison under fixed conditions, not arbitrary domain inference or live source truth.

Builder contract

- Purpose: Create attributable bounded state.
- Central claim: Material packet propositions retain provenance and uncertainty.
- Falsifier: Assembly loses a conflict or cannot trace a governing value.
- Implementation artifact: Conceptual packet schema
- Concrete example: Fresh reads still need compatible governing versions.

Chapter 10 - Make Constitution Status Explicit

Complete constitution does not mean favorable facts. This distinction should be visible in the interface, the code, and the operator's vocabulary.

A pack can completely establish that a recipient is suppressed. All required observations may be available, current, attributable, and consistent. Constitution is sufficient. The adverse value remains part of the state Harmonic evaluates. Turning complete into permitted would substitute evidence availability for present standing.

Conversely, a missing preference source does not establish that suppression is false. The packet should record a missing required proposition, the source expected to establish it, and a remediation path. Distinguish stale, absent, malformed, contradictory, and unsupported-source evidence because each needs different work.

Do not force multiple failures into one unexplained status. Retain a findings list even when the outer interface requires a summary. A packet can contain stale delegation evidence and unresolved preference conflict at the same time. Repairing one dependency does not repair the other.

The executable demonstration reports CONSTITUTED or UNRESOLVED with bounded evidence findings. It does not return PRESERVED, DEFEATED, or a permit. This makes a simple examination possible: change a current suppression value from false to true. Constitution remains complete; the proposition changes. If the constitution code instead returns deny, it has taken on the wrong responsibility.

Give operators actionable constitution findings

An unresolved packet should identify the proposition, expected source, defect, scope, and competent remediation path. 'Unable to send' tells the operator almost nothing. 'Required preference observation is absent for R1 under campaign C4' points to a specific evidence dependency.

Do not expose sensitive underlying evidence to every operator merely to make the error detailed. A restricted reference may support a useful

explanation: which proposition is missing or conflicting, who can inspect the relevant record, and what procedure can establish it. Access control and recourse should work together.

Treat multiple findings independently. If delegation is stale and preference sources conflict, refreshing delegation leaves the conflict unresolved. A status transition should reflect the current findings rather than a user's dismissal of the first notification.

What a constitution status cannot tell you

CONSTITUTED means the declared assembly requirements were met under the examined rules. It does not mean the world is favorable, the institution selected every material predicate, or a provider is permitted to act. CONSTITUTED_WITH_PRECEDENCE adds that a declared hierarchy resolved disagreement, with retained evidence.

The two complete states remain compatible with adverse domain relationships. A comparison can be completely constituted and establish a contradiction. Harmonic then evaluates that relationship for the exact consequence. The pack's state should let a reviewer see the available grounds without anticipating the final disposition.

An outer summary may simplify display, but the findings and resolution records remain first-class artifacts. Keep them versioned and inspectable so later review can distinguish good uncertainty from a defective adapter.

Builder contract

- Purpose: Separate evidence sufficiency from standing.
- Central claim: Adverse-but-complete state remains completely constituted.
- Falsifier: Complete becomes permission or missing evidence becomes a favorable value.
- Implementation artifact: Constitution status and findings model
- Concrete example: Suppression true is complete evidence, not missing evidence.

Chapter 11 - Hand Off to Harmonic Without Answer Leakage

Everything Harmonic legitimately needs should go in. The case-specific answer should not.

That rule does not require stripping the packet of institutional meaning. Source competence, applicable purpose restrictions, scoped approvals, and precedence rules are legitimate governing information. Their origin and applicability must be attributable. A request to 'permit this case because it is important' is different: it supplies the desired conclusion without establishing its authority.

Answer leakage can hide in prose as well as fields. Removing `preferred_outcome` is insufficient if a narrative says the present action must be accepted regardless of contrary evidence. Conversely, a genuine signed approval is not leakage merely because its wording authorizes an action. The distinction is whether the packet preserves an independently established authority instrument or fabricates a case answer.

Keep examination expectations separate from production inputs. A specimen manifest may say that a reviewer expects defeated standing after a valid withdrawal. The submitted packet should contain the withdrawal and applicable governing rule, not the expected label. Blind comparisons should prevent the evaluator from seeing the answer key.

A key scanner catches obvious defects; it cannot certify semantic neutrality. Review provenance and instruction-bearing text. In a production adapter, also verify how unresolved findings, consequence identity, and rule versions map into the actual interface. Never invent a wire contract because the conceptual packet looks tidy.

The inspected single-call interface

The supplied repository identifies `POST /api/evaluate` as the single-call interface. Enabled service determines whether evidence-bearing governance and the secure-execution stage are available. A transport success does not replace inspection of governance, assurance, and execution envelopes.

requested_action describes operation and scope. understanding_state supplies the structured understanding witness. state_transition supplies prior reference, new observation references, attributable relationship, and scope. declared_reality and observed_reality retain claims and observations. authority_chain and revocation_state represent authority context and checks. present_state_provenance records attribution. Obligation and consequence witnesses establish their own requirements.

These are field families, not a sufficient production template. Different domains need different legitimate values, and the runtime supports representations not exhaustively described by the static schema. The v0.3.0 bridge evaluates preserved packets literally instead of generating requests from generic booleans.

Never manufacture a missing witness

The bridge removes understanding_state from a preserving specimen. The runtime records NOT_ESTABLISHED and refuses continuation in that examined case. The bridge does not inject current, complete, provenance_preserved, or applicable to repair it.

Translate representation when source meaning is established. Do not turn absence into positive authority or add verification timestamps because transport occurred now. Source verification time and request time establish different propositions.

A separate controlled specimen introduces an unresolved understanding tension. The result is CONSTRAINED: admissible is true while should_execute is false. An adapter must read the execution boundary rather than reducing the response to one convenient boolean.

Operational outcomes and standing concepts

The inspected runtime uses PERMITTED, CONSTRAINED, ESCALATED, REFUSED, and EMERGENCY_CONTINUITY. They do not create a universal one-to-one mapping onto preserved, defeated, and unresolved standing. Refusal can follow affirmative contradiction or missing required evidence. The latter may remain unresolved rather than defeated.

Preserve reasons and controlling findings. The bridge returns observed outcome, admissible, action, boundary, assurance, and execution fields

without inventing a standing label. A product interpretation needs its own declared rule and must preserve the underlying distinctions.

Record an alignment finding instead of inventing defaults

OpenAPI requires `prior_state_status` and `revalidation_required` for `StateTransition`. The preserved V113 packets omit both and use `consistent_with_prior_state`, absent from its listed enum. The supplied runtime still discriminates the pair. This is a contract alignment finding, not permission to insert current or false values merely to satisfy a schema. Unsupported defaults could supply governing meaning and corrupt predecessor lineage.

Builder contract

- Purpose: Transmit grounds without the desired answer.
- Central claim: Case-specific expected results remain outside determination inputs.
- Falsifier: An outcome instruction changes evaluation without changing legitimate grounds.
- Implementation artifact: Handoff contract and leakage checklist
- Concrete example: Expected DEFEATED is in the examiner manifest, not the packet.

Chapter 12 - Bind Determination to Execution

A standing determination governs the effect only when the component capable of causing that effect obeys it.

The boundary must verify the deployed contract's authenticity, validity, intended executor, material consequence identity, and applicable invalidation state. Binding includes all fields whose substitution would change the governing case. A message version bound before rendering may be inadequate if later enrichment changes the actual disclosure.

The interval between determination and effect remains a race window. Define the mechanism used to constrain it: executor-side version checks, short validity, transactional coupling, or another justified method. Measure relevant invalidation propagation. Do not describe a bounded race as eliminated merely because a timestamp exists.

Retries add a separate question. If the provider may already have accepted the effect, a fresh standing determination does not establish that replay is a new justified consequence. Reconcile provider state and preserve the stable effect identity. Transport ambiguity and standing uncertainty are different conditions even when both require a hold.

Finally, examine causality. If an independent fraud hold would have stopped a payment, a governance refusal plus no payment does not establish that governance prevented it. In a controlled examination, make the provider otherwise executable, intervene on the governing boundary, and observe whether execution depends on that control. Report determination correctness, boundary enforcement, and causal efficacy separately.

Trace the last material mutation

An outreach pipeline may choose a template, personalize the body, attach a brochure, resolve the destination, and invoke a provider. A standing determination made before these operations can govern the final send only if its scope actually covers the resulting material fields. Trace each transformation and decide where the consequence becomes concrete enough to evaluate.

Some transformations may be bounded by an approved rule rather than a single immutable body. That is an institutional choice that must be represented and examined. 'The renderer always does something reasonable' is not a payload identity specification. A changed attachment can create a different disclosure even when the subject line remains identical.

At the boundary, compare the proposed effect with the effect the connector will perform. Verify the actual deployed determination contract, intended executor, authenticity, validity, and relevant revocation or version conditions. A matching local hash alone does not establish a valid permit.

A race is a measured interval

Let determination complete, then inject an authoritative revocation before the provider call. Observe the real route: queue delay, replica reads, event delivery, connector behavior, and executor verification. State where invalidation is guaranteed, what propagation was measured, and what outage or excessive age causes a hold.

A short expiration reduces a window; it does not necessarily detect a revocation inside it. A version check is useful only if the version read and enforced belongs to the authoritative state relevant to the effect. Two independent components can each perform their check correctly and still leave an interval the institution has not justified.

Use explicit bounded claims. 'The tested worker refuses dispatch when required evidence exceeds this age' is examinable. 'Every consequence always reflects the latest reality' is not supported by a timer. Report the remaining interval and the conditions under which the claim holds.

Retry is a consequence question

After a provider timeout, distinguish confirmed nonacceptance from possible acceptance. If the provider accepted payment, repeating it may create another payment. Reconciliation should use the stable effect identity and provider evidence. A new determination of present standing cannot establish that the earlier effect did not occur.

Keep provider state separate: not attempted, confirmed accepted, confirmed failed before acceptance, or ambiguous. Keep governing state separate as well. Both may require a hold for different reasons. Do not ask the operator to grant new authority when the actual missing information is whether the provider already acted.

Correct refusal, enforced refusal, causal prevention

A correct standing result concerns the governing case. An enforced refusal concerns whether the boundary stopped its governed path. Causal prevention asks whether the consequence would otherwise have occurred in the examined conditions.

Suppose a payment has both defeated standing and an independent fraud hold. No payment occurs. That observation supports neither a claim that the governance boundary was reached nor a claim that it caused prevention. Build controlled tests with the independent blocker absent, the provider otherwise executable, and an observable boundary intervention. Then test with the blocker present and report the causal claim as unresolved where the evidence cannot distinguish the causes.

These examinations should never require performing a harmful real effect. Use controlled environments and faithful integrations. Name which properties the environment preserves and which it does not.

Optional secure execution in the same call

The request may include `secure_execution`, whose intent identifies actor and intent and whose execute object identifies the exact action and payload. Service resolution belongs to configured targets and project scope, not a caller-selected arbitrary destination URL. Forwarding requires a stable idempotency identity in the inspected contract.

The stage requires an execution-admitting governance boundary and governance-evidence references before binding. That requirement does not move domain authority into execution. A `BOUND_NOT_FORWARDED` result is a binding without forwarding.

`EXECUTION_OUTCOME_UNKNOWN` preserves ambiguous observation. `EXECUTED` must be read with executor evidence rather than treated as independent proof of every downstream physical effect.

Receipt checks have their own scope

The inspected verifier checks signature, intended service, time validity, and execute hash. The effect-capable receiver must perform its required checks before acting, using legitimate trust configuration. Examine clock tolerance and invalidation behavior rather than assuming a timestamp detects every intervening revocation.

A signature is not durable one-time consumption. Duplicate prevention also requires effect identity, reservation or consumption behavior, and provider reconciliation. V114's one-effect positive control does not establish that repeated valid requests can never create duplicates.

The local V114 result

The test's synthetic executor advances its effect counter only after verification. The positive control produces one effect. Receiptless, forged, altered-payload, expired, refused, and missing-governance-evidence paths produce none. The rerun returns `SUPPORTED_FOR_EXAMINED_BOUNDARY`.

V114 supplies synthetic evidence objects and disables the persistence requirement in its test process. It therefore does not demonstrate real database writes. It does not enumerate every effect-capable deployment route. Preserve the useful result within the boundary it actually examined.

Builder contract

- Purpose: Tie the decision to the effect and examine causal control.
- Central claim: The governed path requires a valid compatible determination for the actual effect.

- Falsifier: Mutation, expiry, revocation, or another connector bypasses the boundary.
- Implementation artifact: Binding and enforcement examination contract
- Concrete example: Recipient drift rejects reuse; ambiguous acceptance requires reconciliation.

Examine, Release, and Maintain Packs

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 13 - Recognize Pack Design Failures

Most pack failures are ordinary implementation shortcuts wearing authoritative labels.

Prompt-pack theater turns instructions into alleged evidence. Latest-record wins mistakes recency for competence. Source monoculture treats one database as authority for everything. Policy laundering promotes an internal recommendation into an institutional rule. Missing-source optimism changes unknown into false so a workflow can finish.

Other failures appear at the boundary. Unscoped approval loses actor, purpose, target, limits, and duration. Executor substitution expands a determination to a different payload. Case-specific hardcoding passes a demonstration because its expected result was encoded in the input. Stale-but-authoritative evidence gets the source right and the observation wrong.

Build defective counterparts, not just a list of warnings. Replace the preference registry with a fresh CRM note. Remove the message version from an approval. Hide a conflict behind a selected value. Ask which check detects each defect and what evidence supports that detection. If no check exists, the release must not claim that failure class is covered.

A useful incident review identifies the boundary crossed and the information lost. 'The AI made a mistake' is rarely enough. Was source competence missing? Did interpretation alter scope? Did the executor accept drift? The repair belongs where the invariant failed.

Repair defects where their meaning was lost

Take a missing-source failure. A developer replaces a failed query with an empty preference object. The next step treats the absent suppression flag as false. The defect began at evidence normalization, not in the final message renderer. Repair the mapping so absent evidence retains an insufficiency finding, then examine the queue path that previously consumed the optimistic default.

For latest-record wins, repair the source register and resolver together. A comment saying 'use the authoritative source' is not enough when the implementation still sorts by timestamp. Preserve conflicting claims and record the precedence rule that applies.

For answer leakage, inspect the derivation. Removing `expected_standing` from JSON does not repair a relationship function that looks up the desired test outcome. Replace that lookup with the declared domain rule over attributable facts and keep the examiner's expectations outside the function.

Pair every repaired failure with a useful counterexample

A fix that blocks every send would prevent the original mistaken dispatch but destroy the product. Pair the missing-source specimen with a complete current preference observation. Pair conflict handling with a legitimately resolved hierarchy. Pair revocation with a durable current grant. Pair payload drift with an unchanged exact effect.

These pairs make the claimed distinction visible. The builder can show that the implementation detects the defect and still admits adequately constituted cases for determination. Safety theater often tests only that the stop button works; useful governance also examines the path that should continue.

Builder contract

- Purpose: Make common defects observable.
- Central claim: Declared failure coverage has an examination specimen.
- Falsifier: An injected covered defect passes unnoticed.
- Implementation artifact: Anti-pattern catalog and paired specimens
- Concrete example: A fresh CRM note replaces required preference evidence.

Chapter 14 - Build Preserved, Defeated, and Unresolved Specimens

A specimen is a controlled governing case, not a story with a preferred ending.

Start with one exact consequence and a declared basis. Produce variants that establish preserved, defeated, and unresolved standing under the stated institutional assumptions. Record why the expected result follows, which proposition changes, and which legitimate evidence establishes that change. Keep those expectations outside the packets.

Then add counterexamples. Durable authority should remain usable under its declared conditions. An unrelated profile change should not defeat a send. Restored authority may create a new legitimate basis. A qualified emergency condition may activate an alternate authority path rather than bypass governance. A pack that interrupts everything has not demonstrated useful temporal governance.

Separate levels of examination. The local constitution tests can establish that missing evidence is exposed or an adverse value is preserved. They cannot establish Harmonic's standing result. A determination examination requires a real, verified evaluator interface. An enforcement examination requires the actual provider path or a controlled integration faithful to it.

Mutations are most useful when they isolate one material difference. If a specimen simultaneously changes recipient, purpose, and authority, a surprising result is difficult to interpret. Build small variations first, then interaction cases. Retain seeds, versions, inputs, and observations so another reviewer can inspect the same claim.

Build mutations around one governing proposition

Hold the actor, authority, action, consequence, pack rule, and runtime fixed. Change one standing-bearing domain fact. This makes the relationship between the mutation and observed result easier to inspect than a scenario in which everything changes at once.

The relationship successor uses two typed facts. In the first case, authority A's execution region equals authority B's accepted region. In the second, only B's value differs. The same comparison rule derives the relationship. The same explicit institutional witnesses remain available. Expected PERMITTED and REFUSED results stay in the examiner, outside the domain pack.

Then examine constitution defects independently. Remove a fact. Substitute an undeclared source. Introduce equal-precedence disagreement. Make a required observation stale. Add answer-bearing metadata. These cases should prevent adequate constitution in this reference, without pretending that absence proves a contradiction.

Keep a specimen manifest useful to another reviewer

Record the exact input, constitution version, frozen rule, controlled difference, expected result and rationale, actual result, and examined boundary. Retain source and adapter versions. Identify whether the observation is historical, local replay, a controlled executor, or live operational evidence.

A successful rerun is a successor observation. It does not edit an earlier failed record. If the mutation revealed a missing predicate, expand the declared claim and examine the revised pack as a new version. Nothing gets to inherit completeness that its examination did not establish.

Builder contract

- Purpose: Construct controlled cases and positive counterexamples.
- Central claim: Specimens isolate material effects without leaking expected answers.
- Falsifier: Expected labels appear in inputs or benign changes cause unexplained interruption.
- Implementation artifact: Versioned specimen suite
- Concrete example: Current grant, withdrawal, outage, and benign-title change.

Chapter 15 - Version the Constitution Without Rewriting History

A pack change can alter what counts as sufficient evidence. That makes release management part of governance.

Version the predicate register, source map, precedence rules, freshness rules, transition interpretation, packet schema, and adapter dependencies. A release manifest identifies which versions operated together. A constitution fingerprint helps detect changed content but does not attest that the institution authorized the release.

Changing a field name and changing source competence are different changes even when both fit in a small commit. Classify compatibility by semantic effect. If a new release requires a preference source that the previous version never collected, historical packets cannot silently be treated as complete under the new rules. Record migration and re-evaluation requirements explicitly.

Preserve historical meaning. Replay should use retained inputs and the historical constitution when the question is what the system evaluated then. A current evaluation should use the currently applicable constitution and current evidence. Replaying a formerly preserved case does not renew its authority now.

Exact output reproducibility may also depend on the determination service version and behavior. Retaining a packet guarantees neither identical future output nor legitimacy of the original decision. Retain the original determination and deployed dependencies. Report separately whether reconstruction, reevaluation, and outcome reproduction were actually achieved.

Identify the runtime as well as the pack

Record the inspected runtime contract, adapter version, relevant source hashes, and examination evidence. A package version may span successive commits. An archive comment identifies source metadata; it does not prove what is deployed.

The companion keeps private engine source external and retains the public synthetic V113 packets with extraction provenance. Field values are unchanged; whitespace is reformatted. Replaying against the supplied snapshot is a new local observation, not a rewrite of historical V113 or a renewal of old authority.

A historical examination time supports comparison and must never become current execution time. Material changes to runtime, vocabulary, domain rule, or adapter transformation create successor examinations whose results cannot be inherited from a predecessor.

Builder contract

- Purpose: Retain semantic lineage across releases.
- Central claim: New versions do not silently alter earlier evaluation meaning.
- Falsifier: An update rewrites prior completeness or standing without a migration record.
- Implementation artifact: Release and replay manifest
- Concrete example: A new required source changes current evaluation requirements.

Chapter 16 - Conduct Pack Examination

Examination should produce a bounded evidence report, not a reassuring badge.

Begin with the claim being tested: this pack constitutes the declared state for these consequence classes using these sources under these assumptions. Inspect predicate adequacy with domain owners. Then test source attribution, freshness, precedence, malformed evidence, conflicts, lineage, and answer leakage. Automated structural checks cover only part of that work.

Examine determination through a verified integration and keep expected results isolated. Examine enforcement by making the provider otherwise capable of acting, injecting incompatible or unavailable determinations, and checking whether the effect occurs. Examine temporal races across the actual queues, replicas, and retry paths. Report remaining intervals and outage behavior.

Causal efficacy needs a separate record. A blocked effect with another independently sufficient blocker does not establish dependence on governance. Controlled interventions can examine whether the control changes the outcome, but that result remains bounded to the tested path and conditions.

Record failures and limitations alongside passes. The six-domain validator does not demonstrate authoritative acquisition, distributed coherence, Harmonic determination, production signing, revocation propagation, provider enforcement, or causal prevention. Its tests demonstrate local constitution behavior against declared synthetic rules. Separate local Harmonic examinations have their own evidence and limits. That is useful evidence when named accurately.

The examination report has four ledgers

The constitution ledger records requirements, source mappings, observation adequacy, interpretation, conflicts, and lineage. The determination ledger records the verified evaluator input and output for the exact case. The enforcement ledger records boundary checks and provider observations. The causal ledger records interventions, independent blockers, and evidence of dependence.

Keeping separate ledgers is not an invitation to collect unlimited logs. It is a way to prevent one kind of evidence from inheriting another kind of claim. A local packet check belongs in the constitution ledger. It cannot be promoted into a claim that Harmonic determined correctly or that an external provider obeyed.

Examination sequence

First, ask domain owners to challenge consequence scope and predicate coverage. Present adverse examples and benign counterexamples. Record which required proposition each example exercises. If an owner cannot identify which source establishes a proposition, hold that claim open rather than letting the adapter select a convenient source.

Second, test packet assembly under missing, stale, malformed, conflicting, duplicate, and incompatible-source observations. Test observation-age boundaries and future timestamps. Mutate material consequence fields while retaining the old evidence scope. Record actual findings and any surviving omissions.

Third, examine a real determination interface with expected answers isolated. Retain request identity, adapter version, evaluator version where exposed, response, and outcome mapping. An interface returning a successful HTTP status does not itself establish preserved standing or executable authority.

Fourth, test execution enforcement through every declared effect path. Include background workers, retry queues, alternate connectors, and final material payload transformations. A missing path is a coverage limit, not an implicit pass.

Finally, introduce change between evaluation and attempted effect. Measure race behavior and include outage cases. Examine causal dependence separately where it matters to the claim. Record operational usefulness through durable-authority and benign-change counterexamples.

Publish failures in useful form

An examination report should tell a builder what failed, what consequence was affected, what evidence was absent or misinterpreted, which rule or component controlled the behavior, and how the claim must be narrowed or repaired. 'Fail' without lineage is almost as difficult to act on as an unsupported pass.

Do not erase a failed specimen after a repair. Preserve the original packet, deployed version, failure observation, repair, and later examination result. The lineage should show what was established at each stage. A current success does not retroactively turn the earlier implementation into a success.

Name the boundary of each result

The companion contains three kinds of evidence: local constitution specimens, selected examinations against a supplied Harmonic source snapshot, and two synthetic derived-relationship handoffs. The source and evidence notes identify the reports and their limits. The six domain examples do not inherit the results of the V113 bridge.

A local validator pass establishes the behavior checked under declared fixture rules. A successful evaluator handoff establishes an observed response for that request and implementation. A synthetic executor test establishes behavior in that tested execution path. A live provider observation and a causal intervention would establish different things. Record each result in its own ledger and leave unexamined layers open.

Builder contract

- Purpose: Produce a bounded examination report.
- Central claim: Each claimed capability is supported at its own examination layer.

- Falsifier: A pass is generalized to untested sources, paths, or causal claims.
- Implementation artifact: Pack Examination Protocol
- Concrete example: An unavailable required source yields a constitution finding.

Build Reference Packs

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 17 - Communications and Outreach

The reference consequence is one promotional send: message M17, campaign C4, institution I1, sender S1, recipient R1 at the verified destination, through provider P1. The actual rendered message and attachments must remain within the evaluated identity.

The synthetic institution declares ten predicates: recipient identity, promotional preference, suppression, purpose, campaign approval, content approval, sender delegation, account eligibility, current reply state, and quota reservation. Each maps to a competent source. Preference and suppression share a registry; sharing a source does not merge their propositions. The versioned constitution records freshness requirements and evidence findings.

For the baseline, every declared proposition is available and current. A reviewer may expect preserved standing under the illustrative rules. In the withdrawal specimen, suppression is authoritatively true: constitution remains complete while the expected standing changes. In the outage specimen, preference evidence is absent: the pack exposes the missing dependency. These expectations live in a separate examiner manifest.

The local implementation verifies source mapping, observation age, provenance fields, missing dependencies, declared conflicts, and obvious answer-bearing keys. It neither contacts the source systems nor calls Harmonic. Its integrity hash binds the synthetic consequence representation; it does not sign a permit. This separation lets a reader inspect what the pack actually does.

The production examination target is larger: prove acquisition from legitimate sources, demonstrate reply and suppression changes reaching queued workers, bind the final provider payload, and test retries after possible acceptance. The reference directory is a starting implementation artifact, not evidence those production properties have been achieved.

Walk the send from schedule to boundary

At scheduling, the campaign worker records message M17 for recipient R1. Scheduling creates an execution proposal, not a durable permit. At dispatch, the worker must constitute the relevant current state for that exact send. The content version, represented institution, purpose, verified destination, campaign, and provider remain explicit.

The illustrative quota ledger reports a reservation for send-001. That proposition is stronger than 'the daily count looked low ten seconds ago.' A production implementation would have to establish reservation ownership, current validity, consumption behavior, and concurrency safety. The synthetic pack records the proposition but does not implement the ledger.

The reply ledger reports whether a relevant reply exists. Relevance is a domain rule: an automatic out-of-office response may not be equivalent to a human request to stop, while an explicit withdrawal should not be discarded as a generic reply. The pack should preserve the source observation and interpretation rather than hiding both behind a convenient inbox flag.

Suppression is evidence, not an outage

In `withdrawn.json`, the preference registry establishes suppression. Every required observation remains well formed and current. Constitution succeeds. That success says the adverse state is available for determination; it does not authorize the send.

In `missing.json`, preference evidence is absent. The validator names the missing required predicate. It does not insert false suppression or infer permission from an empty result. In `stale.json`, a historical observation remains present but exceeds the declared window. Evidence availability and usable currency remain separate.

In `conflict.json`, contradictory claims are retained. The local implementation has no examined precedence resolver, so it reports unresolved conflict. It does not select the last claim in the list. This is a deliberate feature boundary, not proof that institutions can never resolve the conflict.

What the local examiner checks

The command-line examiner loads the constitution and packet, fixes the examination time from the specimen manifest, and runs local constitution checks. It compares the observed status and finding kinds with the separate examiner expectations. The standing expectation is printed as `NOT_RUN` for integration review rather than passed through to the validator.

Each evidence record now carries the synthetic consequence fingerprint. The drift specimen changes the destination without renewing those evidence declarations. The resulting scope mismatch makes reuse visible. An integrator must still decide how authentic scoped claims are acquired and how the final payload is bound to a valid determination.

Production falsifier

The strongest next integration specimen schedules a send, records an authoritative withdrawal or relevant reply, and allows the queue worker to reach its normal dispatch path. The test then observes whether fresh constituted state reaches Harmonic and whether the provider path requires a compatible determination. It should also test a provider timeout after possible acceptance and a connector that enriches attachments.

Until those observations exist, the pack's operational claim remains local constitution under declared synthetic rules. That scope is useful: it tells the next builder precisely which evidence must be earned rather than inherited.

The communications adapter still needs source mapping

Repository inspection supplies an actual runtime boundary to target. It does not legitimize a direct translation from `promotional_preference` into a universal disposition. Map scoped evidence into declared understanding, authority, obligation, reality, and consequence witnesses through established institutional rules.

A reply may alter an outreach prerequisite; a withdrawal may alter the operative basis; a new attachment may alter consequence identity. Preserve the rule, source, and derivation for each relationship. Never fill

missing fields from the expected specimen result.

The next integration examination should retain source observations, pack output, exact request, response, boundary verification, and provider observation for one send. The V113 bridge supplies a method, not a result the communications adapter can inherit.

Builder contract

- Purpose: Work one pack end to end at the constitution layer.
- Central claim: The declared communications state can be assembled and examined locally.
- Falsifier: Suppression, recipient drift, or required-source failure disappears from the packet.
- Implementation artifact: Communications constitution and synthetic packets
- Concrete example: A reply arrives after scheduling and before dispatch.

Chapter 18 - Payments

Define the payment as principal, represented institution, invoice identity, obligation, amount, currency, destination, provider, and stable effect instance. 'Pay the vendor' loses too much of the actual consequence.

The institution's design pack maps payment authority to the delegation instrument, obligation to the approved invoice and matching record, destination to the verified vendor registry, amount limits to the applicable approval, and restrictions to the declared hold sources. An email requesting new bank details is evidence of a request. It does not acquire the registry's authority by being recent.

Precedence and freshness follow the chosen institution's rules. Do not import a universal sanctions or fraud requirement from this example. If a restriction source is required for the declared consequence, its outage must be represented. A current destination record is still inadequate when its verification scope does not cover the proposed beneficiary.

Use three specimens: unchanged verified destination with applicable authority; authoritative withdrawal or an amount outside the grant; required destination verification unavailable. Add a destination-substitution case and a provider-timeout case. The latter may leave payment standing preserved while execution status remains ambiguous.

The examination target is whether an altered destination or duplicate effect can inherit the original determination. This chapter specifies a design reference. It does not implement banking transport, financial compliance, source attestation, or a production payment boundary.

Configure the payments example

The executable reference declares the exact consequence class `payment_release`. Its baseline contains material fields specific to this domain: `target`, `destination`, `amount`, `currency`, `invoice`, `purpose`, `payload_identity`. These fields participate in the local consequence fingerprint. The synthetic principal, institution, provider, and effect instance are explicit as well.

The proposition-to-source map is:

- `principal_verified`: identity-service; baseline value true; illustrative maximum age 60 seconds.
- `payment_delegated`: delegation-registry; baseline value true; illustrative maximum age 60 seconds.
- `invoice_obligation_current`: invoice-ledger; baseline value true; illustrative maximum age 60 seconds.
- `destination_verified`: vendor-destination-registry; baseline value true; illustrative maximum age 60 seconds.
- `amount_within_grant`: approval-registry; baseline value true; illustrative maximum age 60 seconds.
- `restriction_active`: restriction-registry; baseline value false; illustrative maximum age 60 seconds.

Each proposition is an illustrative scoped claim about the exact represented effect. It is not a substitute for the source instruments that would establish that claim in a real institution. Values such as current, verified, covered, and competent require declared semantics and competent ownership before deployment.

Examine the variants

`baseline.json` completely constitutes favorable evidence. `adverse.json` changes `payment_delegated` while keeping the observations complete. The expected standing differs, but the local validator deliberately returns complete constitution in both cases. This is the main boundary examination: adverse reality does not become an evidence outage, and evidence completeness does not become permission.

`missing.json` removes `destination_verified`. `stale.json` ages a required observation beyond the illustrative limit. `conflict.json` preserves disagreement without an implemented resolver. `wrong-source.json` substitutes an informal source. `malformed.json` supplies the wrong type. Each exposes a different constitution dependency; none receives an optimistic default.

scope-drift.json changes the target while retaining evidence scoped to the earlier effect. Its findings demonstrate representation-level scope mismatch. The test does not prove authentic signing, source truth, or rejection by a provider. A production implementation must earn those properties separately.

Handoff and boundary

Transmit attributable propositions, exact effect identity, applicable institutional rule references, and remaining uncertainty through a verified determination adapter. Keep the examiner manifest outside the input. The eight expected standing labels in that manifest are NOT_RUN targets; they are neither supplied to Harmonic nor reported as actual Harmonic observations.

The execution examination must bind the material effect actually performed and cover alternate paths, intervening change, and ambiguous acceptance. Retain the real determination and provider evidence. A successful local constitution examination cannot substitute for that work.

The falsifier for this reference is acceptance as complete when the declared requirements are missing, stale, conflicting, out of source scope, malformed, or bound to a different effect. A domain expert may also falsify the proposed register by identifying an omitted material predicate. That challenge concerns constitution adequacy rather than a parser defect.

Configure raw facts before reducing them to booleans

The payment example begins with boolean propositions to keep the local constitution checks small. A production pack should retain the material amount, currency, grant ceiling, beneficiary identity, destination version, and obligation relationship behind those propositions.

For example, 'amount within grant' can be derived from an attributable payment amount and a scoped approval limit under a declared currency and comparison rule. Do not compare an amount in one currency with a ceiling in another without an established conversion basis. Do not infer beneficiary equivalence from a display name.

Destination evidence should identify the verified account and the scope of that verification. An updated account may be legitimate, fraudulent, or simply unestablished. The pack records what the authoritative process establishes. Harmonic determines present standing for the actual proposed payment destination.

Preserve reservation and provider state

A budget or payment allocation observed before competing work may not remain available. If the workflow relies on a reservation, record which effect holds it, its limits, and its release or consumption conditions. The synthetic boolean is an illustration of that proposition, not an implemented ledger.

After possible provider acceptance, reconcile the original payment instance. A new authorization does not establish nonpayment of the earlier instance. Keep the provider identifier, observation, and stable effect identity attached to the governing lineage. The useful pack remains selective: it needs the payment-relevant grounds, not every financial fact the institution possesses.

Builder contract

- Purpose: Model payment identity and governing dependencies.
- Central claim: Payment constitution preserves destination, amount, obligation, and authority scope.
- Falsifier: A destination substitution is invisible or blindly replayed after ambiguous acceptance.
- Implementation artifact: Payment design pack
- Concrete example: New bank details arrive after invoice approval.

Chapter 19 - Healthcare Workflows

Choose a bounded workflow consequence: transmit an existing signed order to a named receiving system for a named patient. The pack does not choose treatment or create clinical authority.

The institution declares which patient identity service, order system, credential or delegation source, consent record, and workflow policy are competent for this transmission. If current clinical data is required for this exact step, identify the responsible clinical source and interpretation owner. Do not let a model summary supply an absent order or replace professional judgment.

Constitute current order identity and status, clinician authority, intended patient, receiving target, purpose, and required handoff state. A superseded order remains authentic history. Its historical signature does not establish that it is the current order for transmission. Two unresolved order versions require an explicit conflict record or competent resolution.

The preserving specimen uses an operative signed order and current required evidence. The adverse specimen uses authoritative cancellation. The unresolved specimen uses conflicting patient identity or unavailable required order status. Include an emergency alternate path only if the institution has actually defined its source, competence, scope, and conditions.

The falsifier is transmission under an invented, superseded, mismatched, or insufficiently established basis. This is a governance design example. Clinical suitability, patient-safety assurance, regulatory applicability, and deployed medical-system behavior require their own competent review.

Configure the healthcare example

The executable reference declares the exact consequence class `order_transmission`. Its baseline contains material fields specific to this domain: `target`, `destination`, `patient`, `order`, `purpose`, `payload_identity`. These fields participate in the local consequence fingerprint. The synthetic principal, institution, provider, and effect instance are explicit as well.

The proposition-to-source map is:

- `patient_verified`: patient-identity; baseline value true; illustrative maximum age 60 seconds.
- `order_current`: order-system; baseline value true; illustrative maximum age 60 seconds.
- `clinician_authorized`: clinical-authority-registry; baseline value true; illustrative maximum age 60 seconds.
- `required_consent_established`: consent-registry; baseline value true; illustrative maximum age 60 seconds.
- `handoff_ready`: handoff-registry; baseline value true; illustrative maximum age 60 seconds.
- `order_cancelled`: order-system; baseline value false; illustrative maximum age 60 seconds.

Each proposition is an illustrative scoped claim about the exact represented effect. It is not a substitute for the source instruments that would establish that claim in a real institution. Values such as current, verified, covered, and competent require declared semantics and competent ownership before deployment.

Examine the variants

`baseline.json` completely constitutes favorable evidence. `adverse.json` changes `order_cancelled` while keeping the observations complete. The expected standing differs, but the local validator deliberately returns complete constitution in both cases. This is the main boundary examination: adverse reality does not become an evidence outage, and evidence completeness does not become permission.

`missing.json` removes `order_current`. `stale.json` ages a required observation beyond the illustrative limit. `conflict.json` preserves disagreement without an implemented resolver. `wrong-source.json` substitutes an informal source. `malformed.json` supplies the wrong type. Each exposes a different constitution dependency; none receives an optimistic default.

scope-drift.json changes the target while retaining evidence scoped to the earlier effect. Its findings demonstrate representation-level scope mismatch. The test does not prove authentic signing, source truth, or rejection by a provider. A production implementation must earn those properties separately.

Handoff and boundary

Transmit attributable propositions, exact effect identity, applicable institutional rule references, and remaining uncertainty through a verified determination adapter. Keep the examiner manifest outside the input. The eight expected standing labels in that manifest are NOT_RUN targets; they are neither supplied to Harmonic nor reported as actual Harmonic observations.

The execution examination must bind the material effect actually performed and cover alternate paths, intervening change, and ambiguous acceptance. Retain the real determination and provider evidence. A successful local constitution examination cannot substitute for that work.

The falsifier for this reference is acceptance as complete when the declared requirements are missing, stale, conflicting, out of source scope, malformed, or bound to a different effect. A domain expert may also falsify the proposed register by identifying an omitted material predicate. That challenge concerns constitution adequacy rather than a parser defect.

Choose one clinical workflow step to govern

Order transmission is deliberately narrower than selecting or administering treatment. Name the patient, operative order version, receiving system, clinician authority, purpose, and institutionally required handoff conditions. Domain experts determine what additional evidence is material to that transmission.

If two order versions disagree, preserve both claims and their provenance. A signed older order may be authentic while superseded. A newer draft may be recent while lacking the relevant competence. Source hierarchy and current applicability decide which proposition can be constituted; fluency does not.

The pack's unresolved record should identify the required dependency without telling a general-purpose model to settle a clinical question. A competent reviewer can establish a new or clarified basis through the institution's declared process.

Examine usefulness alongside interruption

Include a current signed order that should proceed under the declared requirements, a cancelled order that changes the governing case, and a required identity or status source that cannot establish current state. Add a harmless metadata correction to distinguish formatting from material order change.

If emergency authority is part of this workflow, constitute it as an explicit alternate path with competent activation, scope, expiry, and review. Do not implement an unbounded emergency flag that bypasses every predicate. This chapter provides governance configuration patterns, leaving clinical judgment and applicable professional requirements with their competent owners.

Builder contract

- Purpose: Constitute a narrow clinical workflow without prescribing care.
- Central claim: The pack exposes institutional transmission requirements without creating an order.
- Falsifier: A superseded order or uncertain patient identity is silently treated as operative.
- Implementation artifact: Order-transmission design pack
- Concrete example: An order is cancelled before transmission.

Chapter 20 - Enterprise Identity and Access

A valid session establishes something important. It does not establish every ground for every use of a resource.

Consider exporting customer records. Define principal, represented organization, resource selection, data fields, destination, purpose, tool, and effect instance. The identity service establishes who acts; the delegation source establishes export scope; the resource authority establishes applicable restrictions; the purpose instrument establishes the allowed use. These propositions need not share one source.

An authenticated contractor may retain a usable session after export delegation ends. Constituting session validity while omitting current delegation creates the wrong evidence boundary. A revoked parent grant can also affect outstanding child delegations according to the institution's declared attenuation and revocation rules.

Use an in-scope current export as the preserving specimen, authoritative delegation revocation as the adverse specimen, and indeterminate resource restriction as the unresolved specimen. Include an ordinary low-risk read under a durable scoped grant to protect against needless reauthorization.

Examine alternate connectors and existing background workers. If the export endpoint is governed but a legacy integration can perform the same effect without the boundary, the pack has not established institution-wide enforcement. Changing the model while preserving the principal and effect contract should not silently expand authority.

Configure the access example

The executable reference declares the exact consequence class `customer_export`. Its baseline contains material fields specific to this domain: `target`, `destination`, `resource`, `purpose`, `payload_identity`. These fields participate in the local consequence fingerprint. The synthetic principal, institution, provider, and effect instance are explicit as well.

The proposition-to-source map is:

- `principal_verified`: identity-service; baseline value true; illustrative maximum age 60 seconds.
- `export_delegated`: delegation-registry; baseline value true; illustrative maximum age 60 seconds.
- `resource_scope_current`: resource-authority; baseline value true; illustrative maximum age 60 seconds.
- `purpose_covered`: purpose-registry; baseline value true; illustrative maximum age 60 seconds.
- `session_current`: session-service; baseline value true; illustrative maximum age 60 seconds.
- `revocation_active`: delegation-registry; baseline value false; illustrative maximum age 60 seconds.

Each proposition is an illustrative scoped claim about the exact represented effect. It is not a substitute for the source instruments that would establish that claim in a real institution. Values such as current, verified, covered, and competent require declared semantics and competent ownership before deployment.

Examine the variants

`baseline.json` completely constitutes favorable evidence. `adverse.json` changes `revocation_active` while keeping the observations complete. The expected standing differs, but the local validator deliberately returns complete constitution in both cases. This is the main boundary examination: adverse reality does not become an evidence outage, and evidence completeness does not become permission.

`missing.json` removes `resource_scope_current`. `stale.json` ages a required observation beyond the illustrative limit. `conflict.json` preserves disagreement without an implemented resolver. `wrong-source.json` substitutes an informal source. `malformed.json` supplies the wrong type. Each exposes a different constitution dependency; none receives an optimistic default.

scope-drift.json changes the target while retaining evidence scoped to the earlier effect. Its findings demonstrate representation-level scope mismatch. The test does not prove authentic signing, source truth, or rejection by a provider. A production implementation must earn those properties separately.

Handoff and boundary

Transmit attributable propositions, exact effect identity, applicable institutional rule references, and remaining uncertainty through a verified determination adapter. Keep the examiner manifest outside the input. The eight expected standing labels in that manifest are NOT_RUN targets; they are neither supplied to Harmonic nor reported as actual Harmonic observations.

The execution examination must bind the material effect actually performed and cover alternate paths, intervening change, and ambiguous acceptance. Retain the real determination and provider evidence. A successful local constitution examination cannot substitute for that work.

The falsifier for this reference is acceptance as complete when the declared requirements are missing, stale, conflicting, out of source scope, malformed, or bound to a different effect. A domain expert may also falsify the proposed register by identifying an omitted material predicate. That challenge concerns constitution adequacy rather than a parser defect.

Delegation has an upstream and downstream boundary

A parent service can delegate only within its delegable scope. Record the child principal, authorized action, resource scope, purpose, validity, and relevant parent instrument. A child implementation becoming more capable does not enlarge the grant.

Parent revocation must have declared consequences for outstanding child work. It may invalidate an unexecuted delegated permit, require refreshed evidence, or activate another explicitly authorized procedure. Define those semantics before queues accumulate historical authority.

An authentication token should not carry a claim broader than its issuing authority. A current session may establish identity while leaving export

purpose or resource restriction unestablished. The pack constitutes those separate propositions instead of treating login as universal permission.

Inventory the resource effect paths

List the export endpoint, reporting worker, administrative download, scheduled integration, and other effect-capable paths for the defined export class. Some may be outside the pilot's claimed boundary. Say so explicitly and examine their relationship before claiming deployment-wide enforcement.

Changing the model should leave the authority envelope attached to principal and effect. A durable low-risk read under a scoped grant should also remain distinguishable from an export. This protects both useful automation and legitimate authority from accidental expansion.

Builder contract

- Purpose: Separate authentication from effect-specific authority.
- Central claim: Current identity does not replace resource, purpose, and delegation predicates.
- Falsifier: An authenticated session masks revoked export authority.
- Implementation artifact: Access and delegation design pack
- Concrete example: A contractor session survives export-delegation revocation.

Chapter 21 - Procurement and Vendor Operations

A purchase order is a commitment with material terms, not a generic approved object.

Define vendor, purchase-object version, quantity, price, currency, delivery requirements, budget authority, approver scope, contract state, provider, and effect instance. Map each proposition to the source the institution has actually empowered. The vendor registry's active flag may establish onboarding status without establishing banking instructions or delivery suitability.

Suppose an approver accepted a quantity and price under version PO8. The supplier later changes delivery terms. Whether that change matters depends on the approved scope and the consequence. Preserve the original instrument and identify the changed predicate. Do not update the purchase record in place and keep the approval flag as if nothing changed.

The preserving specimen retains the approved object and current applicable authority. The adverse specimen exceeds an approval limit or relies on a withdrawn approval. The unresolved specimen lacks a required budget or contract observation. A benign formatting correction should remain distinguishable from a changed commitment.

The boundary must bind the purchase version actually issued. The pack cannot establish that property from a pre-check alone. Examine final rendering, connector enrichment, and alternate issuance routes. This chapter produces a design pack; actual purchasing-system integration remains a separate implementation and examination task.

Configure the procurement example

The executable reference declares the exact consequence class `purchase_issue`. Its baseline contains material fields specific to this domain: `target`, `destination`, `purchase_object`, `amount`, `currency`, `purpose`, `payload_identity`. These fields participate in the local consequence fingerprint. The synthetic principal, institution, provider, and effect instance are explicit as well.

The proposition-to-source map is:

- `vendor_active`: vendor-registry; baseline value true; illustrative maximum age 60 seconds.
- `approver_authorized`: approval-registry; baseline value true; illustrative maximum age 60 seconds.
- `budget_reserved`: budget-ledger; baseline value true; illustrative maximum age 60 seconds.
- `object_version_covered`: purchase-repository; baseline value true; illustrative maximum age 60 seconds.
- `contract_current`: contract-repository; baseline value true; illustrative maximum age 60 seconds.
- `approval_withdrawn`: approval-registry; baseline value false; illustrative maximum age 60 seconds.

Each proposition is an illustrative scoped claim about the exact represented effect. It is not a substitute for the source instruments that would establish that claim in a real institution. Values such as `current`, `verified`, `covered`, and `competent` require declared semantics and competent ownership before deployment.

Examine the variants

`baseline.json` completely constitutes favorable evidence. `adverse.json` changes `approval_withdrawn` while keeping the observations complete. The expected standing differs, but the local validator deliberately returns complete constitution in both cases. This is the main boundary examination: adverse reality does not become an evidence outage, and evidence completeness does not become permission.

missing.json removes contract_current. stale.json ages a required observation beyond the illustrative limit. conflict.json preserves disagreement without an implemented resolver. wrong-source.json substitutes an informal source. malformed.json supplies the wrong type. Each exposes a different constitution dependency; none receives an optimistic default.

scope-drift.json changes the target while retaining evidence scoped to the earlier effect. Its findings demonstrate representation-level scope mismatch. The test does not prove authentic signing, source truth, or rejection by a provider. A production implementation must earn those properties separately.

Handoff and boundary

Transmit attributable propositions, exact effect identity, applicable institutional rule references, and remaining uncertainty through a verified determination adapter. Keep the examiner manifest outside the input. The eight expected standing labels in that manifest are NOT_RUN targets; they are neither supplied to Harmonic nor reported as actual Harmonic observations.

The execution examination must bind the material effect actually performed and cover alternate paths, intervening change, and ambiguous acceptance. Retain the real determination and provider evidence. A successful local constitution examination cannot substitute for that work.

The falsifier for this reference is acceptance as complete when the declared requirements are missing, stale, conflicting, out of source scope, malformed, or bound to a different effect. A domain expert may also falsify the proposed register by identifying an omitted material predicate. That challenge concerns constitution adequacy rather than a parser defect.

An amendment is a new applicability question

Record what the approver actually accepted: vendor, object version, quantity, price, delivery terms, and any declared amendment tolerance. A later purchase edit does not decide whether the earlier instrument still covers the commitment.

Some institutions authorize bounded substitutions, such as equivalent equipment within a declared specification and price limit. If that rule exists, preserve the evidence establishing equivalence and limit compliance. If it does not, the agent's belief that the replacement is a better deal cannot create authority.

A budget reservation is another scoped instrument. Show which purchase owns it, whether competing commitments can consume it, and how an amendment affects the allocation. A dashboard balance alone may not establish an available commitment.

Examine the final issued object

Compare the evaluated purchase version with the object the supplier receives. Connector enrichment, tax calculations, shipping changes, and attachment substitution can alter material terms. Define which are within the authority envelope and which require reconstitution and determination.

Test both changed terms outside approval and a benign formatting correction inside the declared representation. Retain the original approval and each amendment's lineage. A correction to the current record should not erase what the institution actually approved earlier.

Builder contract

- Purpose: Keep approval attached to the commitment issued.
- Central claim: Procurement constitution retains material object identity and limits.
- Falsifier: Changed terms inherit a scope-incompatible approval.
- Implementation artifact: Purchase-order design pack
- Concrete example: Delivery terms change between approval and issuance.

Chapter 22 - Legal and Regulated Submission

Define one submission or disclosure, not 'handle the legal matter.'
Material identity includes document version, represented client or institution, competent representative, target, purpose, submission scope, timing conditions, and applicable rule references.

The institution must identify who establishes instructions, document approval, representative authority, and governing procedure. A draft memo summarizing a rule is not automatically the authority for its applicability. An unresolved deadline or rule-version dependency should be exposed to the competent reviewer rather than settled through confident paraphrase.

Consider a disclosure prepared under a client's instruction. The client changes that instruction before transmission. The approved document remains an authentic artifact, but its use must be examined against the current instruction and scope. A newer instruction can legitimately supersede an older one only when its origin and applicability are established.

Use unchanged operative instructions as the preserving specimen, authoritative withdrawal of disclosure authority as the adverse specimen, and disputed representative competence or conflicting instructions as the unresolved specimen. Bind the document actually submitted, including attachments and material redactions.

This design example does not determine law, filing requirements, or professional obligations for a real jurisdiction. Its examination target is architectural: whether changed instructions, document drift, or missing governing evidence can inherit an earlier basis. Publication should retain that scope and obtain appropriate domain review.

Configure the legal example

The executable reference declares the exact consequence class `document_submission`. Its baseline contains material fields specific to this domain: `target`, `destination`, `client`, `instruction`, `purpose`, `payload_identity`. These fields participate in the local consequence fingerprint. The synthetic principal, institution, provider, and effect instance are explicit as well.

The proposition-to-source map is:

- `document_current`: document-repository; baseline value true; illustrative maximum age 60 seconds.
- `instruction_current`: instruction-registry; baseline value true; illustrative maximum age 60 seconds.
- `representative_competent`: authority-registry; baseline value true; illustrative maximum age 60 seconds.
- `submission_scope_covered`: instruction-registry; baseline value true; illustrative maximum age 60 seconds.
- `procedural_dependencies_established`: procedure-registry; baseline value true; illustrative maximum age 60 seconds.
- `disclosure_withdrawn`: instruction-registry; baseline value false; illustrative maximum age 60 seconds.

Each proposition is an illustrative scoped claim about the exact represented effect. It is not a substitute for the source instruments that would establish that claim in a real institution. Values such as `current`, `verified`, `covered`, and `competent` require declared semantics and competent ownership before deployment.

Examine the variants

`baseline.json` completely constitutes favorable evidence. `adverse.json` changes `disclosure_withdrawn` while keeping the observations complete. The expected standing differs, but the local validator deliberately returns complete constitution in both cases. This is the main boundary examination: adverse reality does not become an evidence outage, and evidence completeness does not become permission.

missing.json removes procedural_dependencies_established. stale.json ages a required observation beyond the illustrative limit. conflict.json preserves disagreement without an implemented resolver. wrong-source.json substitutes an informal source. malformed.json supplies the wrong type. Each exposes a different constitution dependency; none receives an optimistic default.

scope-drift.json changes the target while retaining evidence scoped to the earlier effect. Its findings demonstrate representation-level scope mismatch. The test does not prove authentic signing, source truth, or rejection by a provider. A production implementation must earn those properties separately.

Handoff and boundary

Transmit attributable propositions, exact effect identity, applicable institutional rule references, and remaining uncertainty through a verified determination adapter. Keep the examiner manifest outside the input. The eight expected standing labels in that manifest are NOT_RUN targets; they are neither supplied to Harmonic nor reported as actual Harmonic observations.

The execution examination must bind the material effect actually performed and cover alternate paths, intervening change, and ambiguous acceptance. Retain the real determination and provider evidence. A successful local constitution examination cannot substitute for that work.

The falsifier for this reference is acceptance as complete when the declared requirements are missing, stale, conflicting, out of source scope, malformed, or bound to a different effect. A domain expert may also falsify the proposed register by identifying an omitted material predicate. That challenge concerns constitution adequacy rather than a parser defect.

Document identity includes the disclosure

A filing or disclosure can change through attachments, redactions, recipients, or submission purpose even when the document title is unchanged. Define the actual effect so these material variations cannot inherit an earlier determination invisibly.

Client instructions need attributable origin and scope. A staff note about an instruction may be evidence for review without being the controlling instruction itself. A representative's authority may cover preparation while excluding submission or disclosure. Preserve those boundaries.

Procedural applicability is also a declared dependency. Identify the competent source and reviewer for the relevant rule and version. The pack should not transform a retrieved summary into an institutional determination of law merely because the text appears authoritative.

Preserve review and recourse

When instructions conflict or representative competence is unestablished, name what needs clarification and who may establish it. A deadline does not manufacture authority. An authorized urgent procedure, if one exists, needs its own evidence and scope.

Examine a complete current submission, an authoritative withdrawal, and an unresolved instruction or rule dependency. Then mutate an attachment after determination and inspect the boundary's response. The claim concerns the declared workflow and evidence architecture, not universal legal compliance.

Builder contract

- Purpose: Govern exact submission identity and current instructions.
- Central claim: The pack preserves declared submission requirements without inventing applicability.
- Falsifier: Changed documents or client instructions inherit an earlier incompatible basis.
- Implementation artifact: Submission design pack
- Concrete example: Disclosure instructions change before transmission.

Operate the Institution

Start with the consequence. Constitute only the grounds it legitimately needs.

Chapter 23 - Own, Maintain, and Challenge the Pack

A pack without an owner is a future incident with a version number.

Assign responsibility for consequence definitions, domain semantics, source stewardship, technical maintenance, releases, and unresolved-case review. A technical maintainer can repair an adapter without acquiring competence to change institutional authority. A domain owner can approve semantics without proving the implementation enforces them. Both responsibilities need a traceable release process.

Source migrations deserve the same attention as policy changes. A status that meant 'active' in the old system may mean something narrower in the replacement. Compare proposition scope, interpretation, freshness, and event behavior. Test the transition before declaring the new source equivalent.

Provide recourse. An affected person or operator should be able to identify what consequence was held or performed, what evidence controlled, and who can competently review it. Limit access to sensitive records while preserving enough explanation to correct errors. A review path that exposes nothing missing will produce ceremonial overrides.

The institution's measure is not simply how many effects were blocked. Track unresolved causes and age, false interruption reports, source health, changes reaching the boundary, and examined bypass paths. Preserve incident findings as new specimens where appropriate. The pack constitutes. Harmonic determines. The executor enforces. People remain accountable for the institutional rules that make those roles legitimate.

Own contract alignment

A preserved accepted packet and its documented schema can diverge. Assign an owner for resolving the contract a new adapter targets, with runtime, institutional semantics, and evidence stewardship represented.

Resolution may require documentation, compatible implementation, or a new adapter specification. Preserve evidence that the change retains

meaning. Never repair a frozen historical packet merely to satisfy a current validator.

If a workflow holds because a witness is missing, explain the proposition and source. If an interface changed, identify the affected version and remediation. Useful recourse does not require disclosing private engine internals or unrelated sensitive evidence.

Builder contract

- Purpose: Establish ownership, release control, and recourse.
- Central claim: Material semantic changes and unresolved cases have competent accountable owners.
- Falsifier: A source migration changes meaning without review or an affected party lacks the declared review path.
- Implementation artifact: Ownership charter and operating procedures
- Concrete example: A source migration changes the meaning of active status.

Appendices - Builder Worksheets

A. Define the Consequence

- Effect class and exact operation:
- Acting principal and represented principal:
- Target identity and verified destination:
- Purpose and applicable scope:
- Material payload, attachments, amount, currency, or object version:
- Provider, executor, and stable effect instance:
- Final material transformation before the effect:
- Which substitutions require a new determination, and why:
- Canonical representation and consistency assumptions:

B. Register Each Predicate and Source

- Predicate identifier and exact proposition:
- Why it matters to this consequence:
- Required, optional, or conditionally required; activation rule:
- Competent source, owner, authority basis, and negative scope:
- Acquisition and attestation method:
- Source, adapter, and interpretation versions:
- Evidence reference, observation time, applicability interval:
- Freshness rule, invalidation events, outage behavior:
- Cross-source compatibility requirement:
- Precedence rule, competent clarification path:
- Minimization, access, and retention requirements:

C. Record the Antecedent Basis

- Basis identity and original consequence scope:
- Constituent authoritative instruments:

- Evidence establishing joint operativity:
- Contemporaneously recorded or reconstructed:
- Applicable rules and effective intervals:
- Missing lineage and limitations:

D. Assemble the Conceptual Packet

- Pack identity, version, constitution fingerprint, and governing release reference.
- Exact consequence representation and identity.
- Antecedent basis where required.
- Scoped predicate values and evidence sufficiency.
- Source identities, versions, provenance, observations and applicability.
- Relevant events and predicate transition semantics.
- Conflicting claims, hierarchy resolution records, missing dependencies.
- Constitution status with all findings; no desired standing answer.

This is an authoring model, not the Harmonic wire contract. The synthetic JSON reference implements a deliberately smaller subset.

E. Examine Without Supplying the Answer

- Specimen identity, pack version, and controlled mutation.
- Production-shaped input stored separately from expected result.
- Expected constitution status and expected standing, with rationale.
- Evaluator and adapter version; actual result and evidence.
- Provider otherwise executable for causal enforcement examination.
- Effect observed, blocked, ambiguous, or not attempted.
- Independent blockers and causal efficacy assessment.
- Temporal race interval, invalidation lag and outage behavior.
- Untested assumptions, failures and coverage limits.

F. Release and Operate

- Semantic change and competent approval.
- Source-map and adapter dependency changes.
- Compatibility and migration requirements.
- Historical retention and reconstruction evidence.
- Current re-evaluation requirements; replay does not renew authority.
- Owner for unresolved cases, incident review and affected-party recourse.

G. Reference-Pack Coverage

- Communications: executable local constitution example with eight synthetic packets.
- Payments, healthcare, access, procurement, legal: sixteen-element design references plus executable local constitution examples with eight packets each; standing and provider integration remain unrun.
- All six domains: expected standing and enforcement claims require separate examination.

H. Inspected Handoff Families

- Endpoint: POST /api/evaluate; enabled service determines the returned envelopes.
- Operation: requested_action and scoped exact consequence.
- Understanding: source-grounded understanding_state; do not infer its positive witnesses from syntax success.
- Change: state_transition, with prior and observation references, relationship, attribution, time, and scope.
- Reality: declared_reality and observed_reality, preserving claims and observations.
- Authority: verified accepted authority representation, scope, issuer, verification, and applicable history.
- Revocation: acquired revocation_state and check time, not an invented empty list.

- Provenance: `present_state_provenance` and evidence behind the epistemic representation.
- Obligations: explicit witness or verified accepted representation; not applicable is a claim rather than absence.
- Consequence: `consequence_profile` and safeguards tied to actual effect.
- Optional effect stage: `secure_execution`, intent, service, exact execute object, idempotency identity, and applicable policy.

This crosswalk is not an exhaustive schema or automatically sufficient request. Resolve the documented transition discrepancy for the deployed adapter.

I. Inspect the Response

- Transport result is distinct from governance result.
- Preserve outcome, admissible, action, reasons, and controlling findings.
- Endpoint boundary: `governance.execution_boundary.should_execute`; direct local result: `execution_boundary.should_execute`.
- Assurance identifies whether evidence-bearing governance was supplied.
- Effect-stage status, executed, executor evidence, and persistence state remain separate.
- A signed receipt needs verification at the intended effect-capable receiver.
- Provider acceptance, failure, and ambiguity are distinct from governing standing.

Do not map every REFUSED to defeated standing, admissible to executable, or a bound receipt to a delivered effect.

J. Repository Examination Receipt

- Archive identity, inspected-file hashes, fixture extraction paths, and unchanged field values.
- Exact local request and observed runtime result at explicit fixture time.
- Suite commands, exit results, and retained outputs.

- Standing, persistence, provider, and deployment coverage stated according to what was actually examined.
- Successor identity with predecessor records unchanged.

K. Constitute a Domain Relationship

- Define the exact compared propositions and their relevance to the consequence.
- Identify typed value domains, source competence, versions, observations, and evidence references.
- Declare the comparison rule prospectively with scope and institutional origin.
- Record source precedence and the limits of permitted fallback.
- Preserve all examined conflicting claims and legitimate resolution evidence.
- Expose missing, stale, malformed, or unresolved required facts.
- Derive the relationship from facts without using expected runtime outcomes.
- Assemble the handoff with independently supplied institutional witnesses; do not infer them from the relationship.
- Record the exact request, observed Harmonic result, and separate provider evidence where actually examined.
- Treat a changed implementation or boundary as a successor and preserve its predecessor.

Source and Evidence Notes

These notes distinguish the book's design claims from observations about a particular implementation. The companion project's `architecture/HARMONIC-CONTRACT-CROSSWALK.md` and `architecture/EXAMINATION-COVERAGE.md` provide the detailed mapping. No result below is evidence of live provider delivery or universal domain correctness.

Prior book and architectural position

Does It Still Stand? (supplied first-edition PDF) provides the temporal-standing background. This book develops a construction method for domain evidence. The responsibility split-an institution declares, a pack constitutes, Harmonic determines, and a boundary enforces-is the architectural position used throughout this guide. A design requirement is not an empirical finding that every implementation already satisfies it.

Harmonic snapshot

The inspected archive is `harmonic-main (42).zip`. Its archive comment identifies commit `0f24fb3b5ce0fcb77c586453718c195faa0d11f2`; the package declares version 4.1.0. These identify the supplied material, not a deployed service. The contract crosswalk records the source paths and the difference between the documented StateTransition schema and the preserved V113 request accepted by the inspected runtime. An adapter must resolve the actual deployed contract before relying on that handoff.

V112 is used for the boundary between attributable domain relationship and constitutional disposition. V113 preserves the complete-information discrimination pair; its adjudication is in `evidence/examinations/v113-complete-information-successor/01_READ_FIRST/V113_ADJUDICATION.md`. The companion reference `harmonic/lineage.json` records the exact raw-result extraction paths and hashes. Preserved fixtures retain their historical values and examination time. They are not renewed authority for a present action.

What was examined locally

The six domain constitutions each have eight synthetic specimens. Their 48 constitution results concern the local Python validator. Standing expectations for these specimens remain NOT_RUN. A complete adverse specimen demonstrates evidence assembly, not permission to act.

The retained dist/harmonic-examination-v0.3.0.json records five selected repository suites and four bridge cases. The bridge calls the supplied local source directly. Its observations include a preserving V113 case, a contradictory case, missing understanding, and admissible but constrained continuation. Repository execution-boundary tests concern their synthetic executor; they do not establish a production provider path.

The retained dist/relationship-examination-v0.4.0.json records two synthetic successor handoffs from the typed-fact relationship example. The equality rule, sources, priorities, freshness window, and explicit institutional witnesses are declared fixture configuration. The report preserves facts, constitution, exact request, and observed response. It does not establish independent source truth or generalize equality into a universal standing rule.

The v0.5.0 and v0.6.0 examination reports each rerun the 48 local constitution specimens. Earlier repository and relationship reports retain their original version and scope; editorial revision does not silently make them new observations. The release review records the automated checks performed for this edition.

Questions left open

Authoritative source acquisition, independent truth of observations, distributed coherence, historical joint operativity, production signing and revocation, exhaustive semantic answer-leakage detection, provider enforcement, and causal prevention require additional examinations. Domain experts must assess predicate coverage and governing rules for each institution. These are explicit work requirements, not failures repaired by a local test count.